

10.6 A 150kHz-BW 15-ENOB Incremental Zoom ADC with Skipped Sampling and Single Buffer Embedded Noise-Shaping SAR Quantizer

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Many IoT applications demand ADCs with high resolution, medium bandwidth, and good energy efficiency. Lately, the incremental ADC is drawing rising attention by favoring system integration with its easy multiplexing and simple digital filtering. By combining a low-power SAR with a low-distortion $\Delta\Sigma$ modulator, the zoom architecture achieves high energy efficiency and high resolution simultaneously [1-2]. However, the conventional zoom ADC can only convert quasi-static signals, since the two stages operate sequentially, and its $\Delta\Sigma$ conversion operates at a slow speed. The dynamic zoom architecture performs coarse and fine conversions concurrently, thus being able to convert varying inputs [3-4]. Yet, limited by the low quantization level of the fine $\Delta\Sigma$, it requires a large OSR for the targeted resolution (e.g., 282.25 in [3] and 87.5 in [4]), which restricts the input bandwidth to several tens of kHz. Moreover, its loop filter relies on charge transfer, where each conversion requires dedicated sampling for its residue generation. As a result, in addition to the limited input bandwidth, the largely repeated sampling operation incurs extra power and design challenge for input drivers. The CT zoom architecture features easy driving while still requiring a large conversion cycle of 8192 in [5]. Recently, the emerging noise-shaping (NS) SAR ADC employs the efficient SAR for the multi-bit quantizer, significantly reducing conversion cycles [6-7]. However, without the initial coarse quantization, high loop filter orders are usually required for high-resolution applications, raising significant hardware cost and design complexity.

This paper presents a fully dynamic incremental zoom ADC that realizes 92.5dB SNDR over 150kHz-BW with a low OSR of 8 and a small input capacitor of 0.8pF. To extend the input bandwidth of the zoom architecture, a 2nd-order 7b NS-SAR is proposed as the fine quantizer that greatly reduces the conversion cycles. In contrast to [7], only one dynamic buffer is required to realize an Nth-order loop filter, ensuring good energy efficiency and low design complexity. To ease the driving requirement, a floating inverter amplifier (FIA)-assisted residue extraction scheme is proposed to perform the loop filter update without destroying the residue voltage, thereby allowing the ADC to skip 75% sampling operations. Besides, kT/C noise cancellation is performed, reducing the sampling capacitance by 10 $\times$. A 6b digital-slope (DS) ADC with a 2nd-order digital prediction is proposed as the coarse quantizer that efficiently tracks the input with minimum DAC switching activity. Prototyped in 28nm CMOS, the proposed incremental zoom ADC consumes 160 μ W at 300kS/s, leading to a Schreier FoM of 182.2dB.

The block and timing diagrams of the proposed incremental zoom ADC are shown in Fig. 10.6.1. It consists of a 6b DS coarse quantizer and a 2nd-order 7b NS-SAR backend. The incremental ADC first samples the input V_{in} , followed by the DS ADC for the coarse quantization. The DS output is then used to determine the NS-SAR reference. A 4-LSB inter-stage redundancy is provided by setting the NS-SAR reference range to 5 DS LSBs. The DS and NS-SAR DACs are merged for better matching. An amplifier is employed before the loop filter to reduce backend noise, and prevents destructive charge transfer as in the conventional loop filter of zoom ADCs [3-4], thereby allowing 4 consecutively fine conversions without additional input sampling. As a result, one shot incremental zoom conversion requires 8 sampling operations and 32 conversions.

Figure 10.6.2 depicts the implementation of the proposed fully dynamic incremental zoom ADC. The split-capacitor array is adopted in both coarse and fine stages to ensure a stable common-mode voltage during the conversion. The 2nd-order NS-SAR is implemented with an error-feedback (EF) structure, where the loop filter is implemented using capacitor stacking and a single flipped voltage follower (FVF) buffer. An amplifier is placed after the DAC for both kT/C noise cancellation and residue amplification. The FIA structure is adopted thanks to its simplicity, energy efficiency, and stable common mode [8]. The ADC operation includes 4 phases (Fig. 10.6.2 bottom-left). First, the input is bottom-plate sampled on C_{DAC} . The kT/C noise in time t_1 will be amplified and stored on C_{NC} , which will be cancelled during the critical fine conversions. The kT/C noise cancellation operates in a fully dynamic fashion. As a result, only a small C_{DAC} of 0.8pF is utilized, largely relaxing input driving requirements and reducing reference power. Next, the sampled input is converted by the asynchronous DS ADC for coarse quantization. To ensure fast tracking, a 2nd-order digital prediction scheme is employed in the DS process. For instance, the n^{th} DS conversion will start with an initial guess of $2 \cdot D_{DS}[n-1] - D_{DS}[n-2]$. With the help of the prediction scheme, only 6 DS steps are needed for a targeted bandwidth of 150kHz. Moreover, it also minimizes the DAC switching activity. After completing the coarse quantization, the residue voltage is then amplified and converted by the NS-SAR quantizer.

Once the NS-SAR quantization finishes, the FIA will amplify the residue for the FIR filter update (Fig. 10.6.2 bottom-right). In the k^{th} conversion cycle, the FVF extracts and stores the residue error of $V_{EF}[k]$ on C_{EF1A} and C_{EF1B} . Those two capacitors are stacked in series in the $(k+1)^{\text{th}}$ conversion cycle. Together with the reversely-connected EF capacitor from the $(k-1)^{\text{th}}$ cycle, it realizes $2 \cdot V_{EF}[k] - V_{EF}[k-1]$ for the 2nd-order FIR filter. Placing them at the output of the FIA, the passive summation of the NS-SAR input and FIR filter is naturally realized. Compared to the prior buffer-assisted capacitor stacking NS-SAR [7], only one buffer is required to realize the 2nd-order NS. Moreover, with this architecture, one can easily increase the NS order without requiring extra buffers.

During the ADC operations, the FIA serves as the amplifier for kT/C noise cancellation, the NS-SAR conversion, and residue extraction consecutively, where the design requirements vary significantly. To ensure a high-resolution data conversion while maintaining energy efficiency, a 3-mode reconfigurable FIA is designed, as shown in Fig. 10.6.3. During the kT/C noise cancellation phase, the sampling noise is amplified by A_{KTC} and then stored on C_{NC} . As it is the noise-critical phase, a large reservoir capacitor of 16pF is adopted. The amplifier gain should remain the same for the other 2 phases (A_{CONV} and A_{RE}) for proper noise cancellation. However, due to different loading conditions, the FIA needs to be reconfigured accordingly. Note that the gain accuracy and noise requirements vary in different modes. E.g., the error generated in the residue extraction phase will be preserved in the FIR filter, while the error in the NS-SAR conversion phase will be relieved by noise shaping. The simulation shows that $\pm 10\%$ gain deviation in the error extraction phase (A_{RE}) leads to a maximum of 0.5dB performance degradation. In contrast, the conversions phase gain (A_{CONV}) can tolerate over $\pm 25\%$ change (Fig. 10.6.3, bottom-right). Similar conclusions can be drawn on noise performances. Thus, to ensure the conversion resolution while maintaining a low power consumption, a 9pF reservoir capacitor is adopted for residue extraction, and only 1.5pF is needed for conversion. An extra loading capacitor of 2pF is added during kT/C cancellation and residue extraction phases to lower its noise. Amplifications in the residue extraction and conversion phase consume only 50% and 15% energy respectively, compared to that in the kT/C cancellation phase. A one-time trimming is performed on the reservoir capacitor for gain centering. As a result, the 3-mode reconfigurable FIA satisfies design requirements across all modes while maintaining excellent energy efficiency.

The prototype ADC in 28nm CMOS occupies 0.026mm² (Fig. 10.6.7). Measured results are shown in Fig. 10.6.4 and Fig. 10.6.5. Foreground calibration is used to address capacitor mismatches. Under a 2.5kHz input, the measured SNDR and SFDR are 92.5dB and 107.2dB. Under a high-frequency input of 20kHz (limited by the signal source), the measured SNDR and SFDR are 92.2dB and 106.1dB. The ADC core consumes 160 μ W at 300kS/s (reference 6.3 μ W; FIA 80.8 μ W; FVF 36.3 μ W; digital and switches 36.4 μ W). The SNR/SNDR vs. input amplitude shows a DR of 93.9dB. Figure 10.6.6 summarizes the performance of this work and compares it with prior publications. Among discrete-time ADCs with similar performance (>90dB SNDR), this work has the minimum equivalent sampling capacitance thanks to the kT/C noise cancellation and the skipped sampling operation. By adopting the efficient NS-SAR as the fine quantizer, the conversion cycles are greatly reduced. Therefore, it achieves the highest input bandwidth of 150kHz compared with recent high-performance incremental ADCs. Combining all the merits, this work realizes a superior FoM of 182.2dB while occupying a small area of 0.026mm², making it a suitable candidate for IoT applications.

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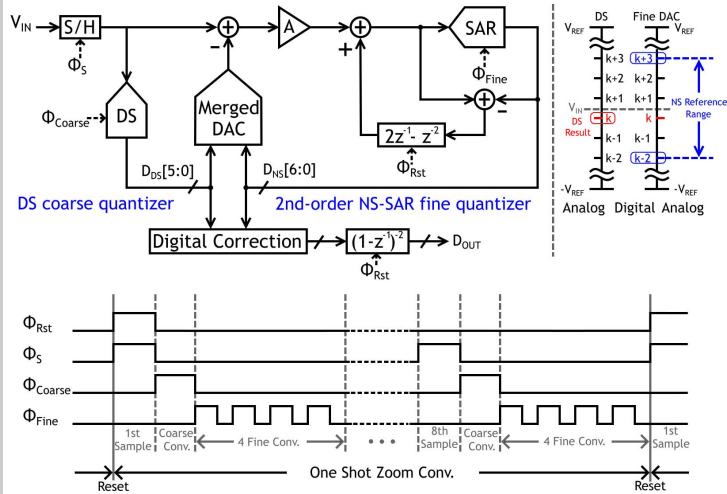


Figure 10.6.1: Simplified block and timing diagram of the proposed incremental zoom ADC.

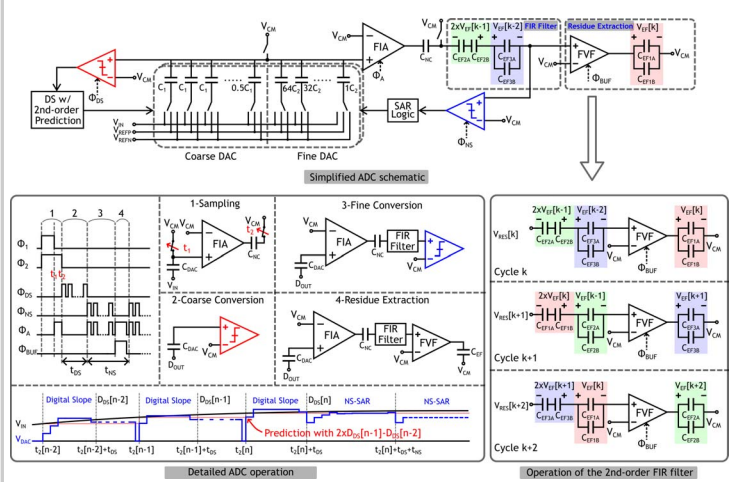


Figure 10.6.2: Simplified ADC schematic (top), ADC operation illustrations (bottom-left), and the 2nd-order FIR filter operation (bottom-right).

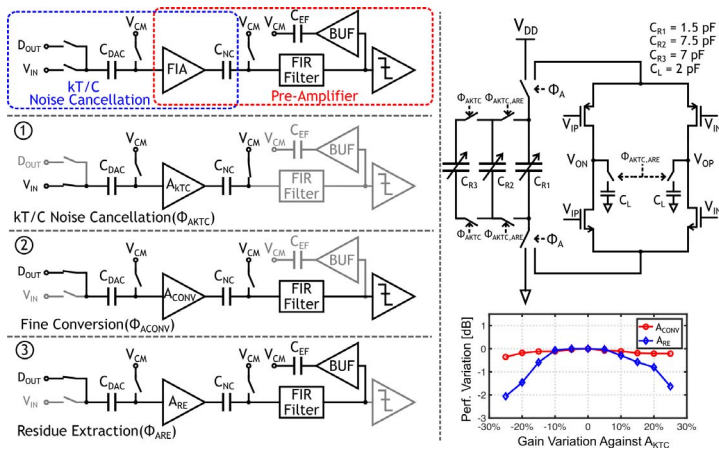


Figure 10.6.3: The 3-mode reconfigurable FIA schematic, operation, and performance variation.

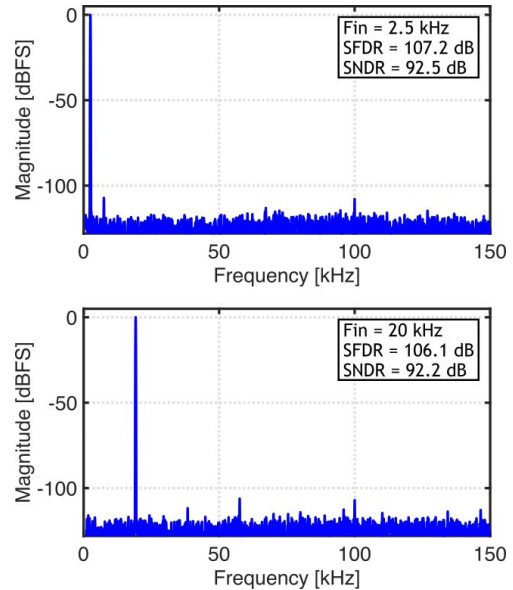


Figure 10.6.4: Measured output spectra with 2.5kHz and 20kHz inputs.

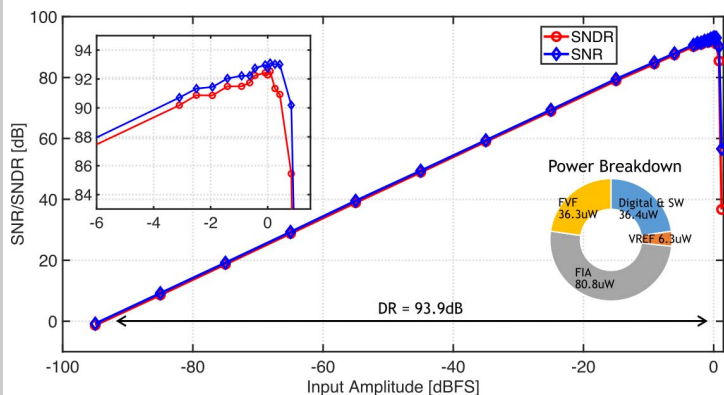


Figure 10.6.5: Measured dynamic range and power breakdown.

	ISSCC-20 Liu	ISSCC-21 Liu	VLSI-20 Eland	ISSCC-22 Jie	ISSCC-22 Liu	This work
Architecture	NS-SAR	NS-SAR	Dynamic Zoom (SAR + $\Delta\Sigma$)	Incremental Zoom (Cnt. + CT $\Delta\Sigma$)	Incremental Zoom (SAR + $\Delta\Sigma$)	Incremental Zoom (DS + NS-SAR)
Process [nm]	40	40	160	28	55	28
BW [kHz]	40	250	20	25	Quasi-static	150
Conv. cycles	--	--	--	8192	128	32
OSR	25	10	87.5	NA ¹	128	8
Sample cap [pF]	18	16	6.8	NA ¹	4	0.8
Equiv. sample cap [pF] ²	450	160	595	NA ¹	512	6.4
Reference [V]	1.1	1.1	1.8	1.2	1	1
Area [mm ²]	0.061	0.094	0.27	0.014	0.23	0.026
Power [uW]	67.4	340	440	590	4.96	160
SNDR [dB]	90.5	93.3	106.5	100.1	93	92.5
FoMw ³ [fJ/c-s]	30.8	18.0	63.7	142.7	50.2	15.6
FoMs ⁴ [dB]	178.2	182	183.1	176.4	177.3	182.2

¹CT front-end omits sampling operation.

²Equivalent sample capacitance = sample cap x OSR, representing the total sampled capacitance for one conversion.

³FoMw = Power / (2^{ENOB} · 2 · BW)

⁴FoMs = SNDR + 10 · log₁₀(BW/Power)

Figure 10.6.6: Performance summary and comparison with state-of-the-art ADCs.

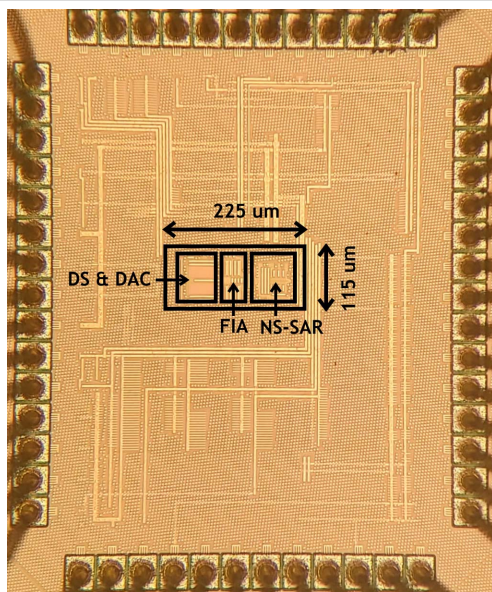


Figure 10.6.7: Chip micrograph.